

Symbiotic BLS12-381 Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Symbiotic on Symbiotic BLS12-381. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated December 11, 2025, was delivered to Symbiotic, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 3 days, the Cyfrin team conducted an audit on the Symbiotic BLS12-381 code provided by Symbiotic. The findings consist of 2 Informational findings & 1 Gas Optimization.

Summary

Project Name	Symbiotic BLS12-381
Repository	relay-contracts
Commit	9b9397f97296...
Audit Timeline	Dec 1st - Dec 3rd, 2025
Methods	Manual Review, Differential Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	0	0	0
Informational	2	1	1
Gas Optimizations	1	0	1
Total	3	1	2

6 Audit Scope

Cyfrin conducted an audit of the BLS12-381 contract for Symbiotic based on the code present in the repository at commit hash 9b9397f. The following file constitutes the scope of the audit:

```
src/libraries/utils/BLS12381.sol
```

7 Risk Classification

	Impact: High	Impact: Medium	Impact: Low
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The BLS12381 library implements the full set of arithmetic and hashing primitives required to operate over the BLS12-381 G1 subgroup, exposing a deterministic and self-contained interface for point construction, mapping, and validation. The contract defines the curve using its affine Weierstrass form over the base field Fp, encoding G1 elements as two-limb field values for both x and y coordinates. The point at infinity is explicitly represented as the all-zero tuple, allowing group operations and precompile interactions to treat it as the identity element.

Hashing a message to the curve follows the structure of the hash-to-curve standard. The message is expanded using SHA-256 in XMD mode, producing a sequence of pseudo-random bytes that are reduced into valid Fp field elements. These elements are then mapped to the curve using the Ethereum precompile for field-to-curve mapping, which performs the Simplified SWU map followed by an isogeny and full cofactor clearing to ensure the resulting point lies in the correct subgroup. The mapped points are added to yield the final hash-to-G1 output.

The library also includes functionality for reconstructing points from x-coordinates using the standard exponentiation-based square-root method valid for primes where p mod 4 = 3. Field operations are expressed in a limb-wise format compatible with EVM word-level arithmetic, including carry handling, modular adjustments, and controlled use of preallocated memory. Group-theoretic operations such as negation, scalar multiplication, and subgroup membership checks are structured around these same primitives and align with the curve's algebraic definition.

Together, these components provide the underlying mathematical operations needed by higher-level key and signature modules, offering deterministic hashing, subgroup-sound point generation, and consistent interaction with Ethereum's pairing and mapping precompiles.

9 Attestation

Cyfrin attests that it performed a security review of the Symbiotic BLS12-381 code provided by Symbiotic.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-12-03-cyfrin-symbiotic-bls12381-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Symbiotic has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=05d3ca1a033d222596ea335a5e4679eba8bc0a84da75464b1e24faa64d7ae13b
F=2025-12-03-cyfrin-symbiotic-bls12381-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.